

GATE MA

Mathematics

TREND-BASED MOCK PAPER

— 02 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	6 subjects

Difficulty Distribution

Easy: 19 questions (29%)

Moderate: 34 questions (52%)

Difficult: 12 questions (18%)

GENERAL INSTRUCTIONS

GATE MA — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
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4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1. 1M [-0.33]

[Algebra]

Ring must have:

- (A) 1 only
- (B) 0 and 1
- (C) Both +, -, ×
- (D) None

Q2. 1M [-0.33]

[Algebra]

Rank of 3x3 identity:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q3. 1M [no -ve]

[Algebra]

Inverse of AB = ?

Enter your answer here

Q4. 1M [-0.33]

[Algebra]

Identity element property:

- (A) $a * e = e * a = a$
- (B) $a * e = e$
- (C) $e * a = e$
- (D) $a * e = a$

Q5. 1M [no -ve]

[Calculus]

Order of $y'' + 3y' + 2y = 0$:

Enter your answer here

Q6. 1M [no -ve]

[Calculus]

Limit of $\sin(x)/x$ as $x \rightarrow 0$:

Enter your answer here

Q7. 1M [-0.33]

[Analysis]

Cauchy sequence:

- (A) Converges always
- (B) Not always
- (C) Never
- (D) Random

Q8.

1M [-0.33]

[Topology]

Union of open sets is:

- (A) Open
- (B) Closed
- (C) Neither
- (D) Both

Q9.

1M [-0.33]

[Probability]

Normal distribution PDF:

- (A) $\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{x^2}{2\sigma^2}\right)$
- (B) $\exp(-x)$
- (C) $1/x$
- (D) x^n

Q10.

1M [no -ve]

[Real Analysis] $f(x)=|x|$ at $x=0$:

Enter your answer here

Q11.

1M [-0.33]

[Algebra]

Rank of 3x3 identity:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q12.

1M [no -ve]

[Algebra]

Identity element property:

Enter your answer here

Q13.

1M [-0.33]

*[Calculus]*Limit of $\sin(x)/x$ as $x \rightarrow 0$:

- (A) 0
- (B) 1
- (C) Infinity
- (D) Undefined

Q14.

1M [-0.33]

[Probability]

Normal distribution PDF:

- (A) $\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{x^2}{2\sigma^2}\right)$
- (B) $\exp(-x)$
- (C) $1/x$
- (D) x^n

Q15.

1M [-0.33]

[Topology]

Union of open sets is:

- (A) Open
- (B) Closed
- (C) Neither
- (D) Both

Q16.

1M [-0.33]

*[Calculus]*Limit of $\sin(x)/x$ as $x \rightarrow 0$:

- (A) 0
- (B) 1
- (C) Infinity
- (D) Undefined

Q17.

1M [-0.33]

[Topology]

Union of open sets is:

- (A) Open
- (B) Closed
- (C) Neither
- (D) Both

Q18.

1M [-0.33]

*[Algebra]*Inverse of $AB = ?$

- (A) $B^{-1} A^{-1}$
- (B) $A^{-1} B^{-1}$
- (C) $(A+B)^{-1}$
- (D) None

Q19.

1M [-0.33]

*[Calculus]*Limit of $\sin(x)/x$ as $x \rightarrow 0$:

- (A) 0
- (B) 1
- (C) Infinity
- (D) Undefined

Q20.

1M [-0.33]

[Algebra]

Ring must have:

- (A) 1 only
- (B) 0 and 1
- (C) Both $+, -, \times$
- (D) None

Q21.

1M [-0.33]

[Topology]

Union of open sets is:

- (A) Open
- (B) Closed
- (C) Neither
- (D) Both

Q22.

1M [-0.33]

*[Calculus]*Limit of $\sin(x)/x$ as $x \rightarrow 0$:

- (A) 0
- (B) 1
- (C) Infinity
- (D) Undefined

Q23.

1M [-0.33]

[Analysis]

Cauchy sequence:

- (A) Converges always
- (B) Not always
- (C) Never
- (D) Random

Q24.

1M [no -ve]

*[Algebra]*Inverse of $AB = ?$

Enter your answer here

Q25.

1M [-0.33]

[Probability]

Normal distribution PDF:

- (A) $(1/\sigma\sqrt{2\pi})\exp(-(x-\mu)^2/(2\sigma^2))$
- (B) $\exp(-x)$
- (C) $1/x$
- (D) x^n

Q26.

1M [-0.33]

*[Algebra]*Inverse of $AB = ?$

- (A) $B^{-1}A^{-1}$
- (B) $A^{-1}B^{-1}$
- (C) $(A+B)^{-1}$
- (D) None

Q27.

1M [no -ve]

[Algebra]

Ring must have:

*Enter your answer here***Q28.**

1M [-0.33]

*[Calculus]*Limit of $\sin(x)/x$ as $x \rightarrow 0$:

- (A) 0
- (B) 1
- (C) Infinity
- (D) Undefined

Q29.

1M [-0.33]

[Topology]

Union of open sets is:

- (A) Open
- (B) Closed
- (C) Neither
- (D) Both

Q30.

1M [no -ve]

[Topology]

Union of open sets is:

*Enter your answer here***Q31.**

2M [no -ve]

*[Algebra]*Inverse of $AB = ?$

*Enter your answer here***Q32.**

2M [no -ve]

[Algebra]

Identity element property:

*Enter your answer here***Q33.**

2M [-0.66]

*[Algebra]*Rank of 3×3 identity:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q34.

2M [no -ve]

[Probability]

Normal distribution PDF:

*Enter your answer here***Q35.**

2M [no -ve]

[Algebra]

Ring must have:

*Enter your answer here***Q36.**

2M [no -ve]

[Algebra]

Rank of 3x3 identity:

*Enter your answer here***Q37.**

2M [-0.66]

[Algebra]

Identity element property:

- (A) $a*e=e*a=a$
- (B) $a*e=e$
- (C) $e*a=e$
- (D) $a*e=a$

Q38.

2M [-0.66]

[Algebra]

Ring must have:

- (A) 1 only
- (B) 0 and 1
- (C) Both +, -, ×
- (D) None

Q39.

2M [no -ve]

[Algebra]

Identity element property:

*Enter your answer here***Q40.**

2M [-0.66]

[Algebra]

Identity element property:

- (A) $a*e=e*a=a$
- (B) $a*e=e$
- (C) $e*a=e$
- (D) $a*e=a$

Q41.

2M [-0.66]

[Algebra]

Identity element property:

- (A) $a * e = e * a = a$
- (B) $a * e = e$
- (C) $e * a = e$
- (D) $a * e = a$

Q42.

2M [-0.66]

[Algebra]

Identity element property:

- (A) $a * e = e * a = a$
- (B) $a * e = e$
- (C) $e * a = e$
- (D) $a * e = a$

Q43.

2M [no -ve]

*[Calculus]*Order of $y'' + 3y' + 2y = 0$:

Enter your answer here

Q44.

2M [no -ve]

*[Algebra]*Inverse of $AB = ?$

Enter your answer here

Q45.

2M [-0.66]

[Real Analysis] $f(x) = |x|$ at $x=0$:

- (A) Discontinuous
- (B) Continuous
- (C) Differentiable
- (D) None

Q46.

2M [-0.66]

*[Algebra]*Inverse of $AB = ?$

- (A) $B^{-1} A^{-1}$
- (B) $A^{-1} B^{-1}$
- (C) $(A+B)^{-1}$
- (D) None

Q47.

2M [-0.66]

[Algebra]

Ring must have:

- (A) 1 only
- (B) 0 and 1
- (C) Both $+, -, \times$
- (D) None

Q48.

2M [-0.66]

[Probability]

Normal distribution PDF:

- (A) $\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{x^2}{2\sigma^2}\right)$
- (B) $\exp(-x)$
- (C) $1/x$
- (D) x^n

Q49.

2M [no -ve]

[Analysis]

Cauchy sequence:

Enter your answer here

Q50.

2M [-0.66]

*[Algebra]*Inverse of $AB = ?$

- (A) $B^{-1} A^{-1}$
- (B) $A^{-1} B^{-1}$
- (C) $(A+B)^{-1}$
- (D) None

Q51.

2M [no -ve]

*[Calculus]*Limit of $\sin(x)/x$ as $x \rightarrow 0$:

Enter your answer here

Q52.

2M [-0.66]

*[Algebra]*Rank of 3×3 identity:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q53.

2M [-0.66]

[Probability]

Normal distribution PDF:

- (A) $\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{x^2}{2\sigma^2}\right)$
- (B) $\exp(-x)$
- (C) $1/x$
- (D) x^n

Q54.

2M [-0.66]

*[Algebra]*Inverse of $AB = ?$

- (A) $B^{-1} A^{-1}$
- (B) $A^{-1} B^{-1}$
- (C) $(A+B)^{-1}$
- (D) None

Q55.

2M [-0.66]

[Real Analysis] $f(x)=|x|$ at $x=0$:

- (A) Discontinuous
- (B) Continuous
- (C) Differentiable
- (D) None

Q56.

2M [no -ve]

[Algebra]

Identity element property:

Enter your answer here

Q57.

2M [no -ve]

[Probability]

Normal distribution PDF:

Enter your answer here

Q58.

2M [-0.66]

[Algebra]

Identity element property:

- (A) $a*e=e*a=a$
- (B) $a*e=e$
- (C) $e*a=e$
- (D) $a*e=a$

Q59.

2M [-0.66]

[Algebra]

Identity element property:

- (A) $a*e=e*a=a$
- (B) $a*e=e$
- (C) $e*a=e$
- (D) $a*e=a$

Q60.

2M [-0.66]

*[Calculus]*Order of $y''+3y'+2y=0$:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q61.

2M [no -ve]

*[Algebra]*Inverse of $AB = ?$

- (A) $B^{-1} A^{-1}$
- (B) $A^{-1} B^{-1}$
- (C) $(A+B)^{-1}$
- (D) None

Q62.

2M [-0.66]

[Topology]

Union of open sets is:

- (A) Open
- (B) Closed
- (C) Neither
- (D) Both

Q63.

2M [no -ve]

[Algebra]

Rank of 3x3 identity:

Enter your answer here

Q64.

2M [-0.66]

[Topology]

Union of open sets is:

- (A) Open
- (B) Closed
- (C) Neither
- (D) Both

Q65.

2M [-0.66]

[Probability]

Normal distribution PDF:

- (A) $\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{x^2}{2\sigma^2}\right)$
- (B) $\exp(-x)$
- (C) $1/x$
- (D) x^n